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RESEARCH

Approach of Genetic Diversity of *Lippia alba* (Mill) and *Petiveria alliacea* L.: Medicinal Plants of Colombia

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Abstract

The medicinal *Lippia alba* and *Petiveria alliacea*, originating from Central and South America, exhibit a wide range of beneficial properties, including antimicrobial, antifungal, anti-inflammatory, antitumor, analgesic, and antibacterial effects. However, little is known about their population structure and genetic diversity, which may hinder the establishment of their cultivation in different regions of Colombia. In this study, we conducted a comprehensive analysis of the genetic diversity and population structure of 17 samples of *L. alba* from the departments of Tolima, Valle del Cauca, and Putumayo, as well as 31 samples of *P. alliacea* from the departments of Cundinamarca, Boyacá, Tolima, and Valle del Cauca. We employed restriction-site associated DNA sequencing (RADseq) with the enzyme PstI. We performed denovo_map and ref_map pipeline for *L. alba* and identified a total of 17,036 loci and 14,562 SNPs, respectively, revealing a genetic variation of 5.19% (FST of 0.05; $p < 0.001$) among its populations and two delimited genetic groups. For *P. alliacea*, in denovo_map our analysis discovered 6395 SNPs, indicating substantial genetic variation of 75% among the studied populations (FST = 0.75; $p < 0.001$), resulting in the delineation of four genetic groups. Our findings will contribute to providing valuable molecular data on the populations of these medicinal plants and provide evidence of the genetic flow existing among *L. alba* populations, while *P. alliacea* populations are more structured. We also performed a FDIST analysis; for *L. alba* using the results with reference, we identified 37 SNPs coding for proteins in biological, molecular, and cellular processes; and finally, we highlighted the gene SASPL_104284 involved in metabolic processes.

Keywords RADseq · PstI · Omics · SNPs · Genetic diversity · Population structure

Key Message
• In this research, we present an approach to genomic analysis of different populations of *L. alba* and *P. alliacea* in Colombia; we found that the first exhibits lower population structure and greater genetic diversity within populations, and the second shows greater structure among populations.

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Introduction

The knowledge of medicinal plants in Colombia can be traced back to traditional practices of ethnic groups and tribes, that have passed down this information from generation to generation. This knowledge is supported by ethnopharmacological studies (Gómez-Estrada et al. 2011) and chemical studies that evaluate the presence of secondary metabolites with pharmacological potential in their extracts (Muñoz-Acevedo et al. 2019). On the other hand, studies on their genetics are scarce, and in some cases, nonexistent. Considering that the study of genetic diversity within and among individuals and species in a population allows us to understand the processes of selection and evolution of populations (Salgotra and Chauhan 2023), as well as serving as a starting point for future studies that seek to understand the mechanism by which genes involved in the synthesis of secondary metabolites function, it is essential to conduct

genomic studies on these plants. The results here obtained could advance our knowledge of these plants and help in selecting superior genotypes that could be used in the establishment of crops aimed at conserving, studying, or utilizing their potential.

To conduct these genetic analyses, different molecular and biotechnological technologies and methodologies have been developed, allowing for an in-depth exploration of the genetic structure and sequence of a species of interest (Scheben et al. 2017). One sequencing (NGS) is Restriction Site Associated DNA Sequencing (RADseq), which can offer valuable information insight into species genomes at a reduced cost compared to complete sequencing. RADseq uses a specific restriction endonuclease to digest DNA into short fragments, enabling de novo analysis that takes advantage of the data's reduced complexity to discover reference-free variants and genotype them (Peterson et al. 2014). This technique allows for the identification of Single Nucleotide Polymorphisms (SNPs), which are considered abundant and high-density genetic markers, making it an ideal tool for evaluating and understanding the genetic and structural mechanisms of genomic regions in different species and their populations (Baird et al. 2008).

Currently, the interest in studying medicinal plants has increased due to their use as therapeutic strategies for various diseases such as cancer. In Colombia, the GAT program (Generation of therapeutic alternatives in cancer from plants) aims to study the genetic diversity of different species that could have antitumor or anticancer activity, such as *Lippia alba* (Mill) and *Petiveria alliacea*, natives of Tropical and Subtropical America, that in Colombia they can be found in domesticated wild or environments throughout all geographic regions. *Lippia alba* (Mill.) N.E.Br. ex Britton and P. Wilson (Lamiales: Verbenaceae), known by common names in Spanish of "Pronto alivio," "Alivia dolor," and "Curalo todo" (Bernal et al. 2017), is a subshrub or shrub and grows primarily in the seasonally dry tropical biome, ranging from 400 to 2120 masl, has a maximum height of up to 2 m, and has pedunculated lilac or white inflorescences. It is categorized as "Least Concern" (LC) by The International Union for Conservation of Nature (IUCN) and on Colombia's National Red List (Muñoz et al. 2007; ColPlantA 2023). Natural populations of this species are allogamous, self-incompatible, and traditionally reproduced by cuttings (Venâncio et al. 2016). A total of eighteen metabolites have been identified, with key compounds including monoterpenes, sesquiterpenes, essential oils, carvone, limonene, flavonoids, alkaloids, phenols, tannins, and saponins (Ministerio de Protección Social 2008); these compounds can be present in the essential oils of the plant and which vary significantly according to the plant's ploidy level (Lopes et al. 2020) which has been described as an autopolyploid complex with five cytotypes (2n = 30, 38, 45, 60, and 90) (de Souza et al. 2021). Using flow cytometry, it has been described as having a genetic size of 2,234 Mb (Leitch et al. 2019) and genome annotation (ASM3783412v1) at the scaffold level with a size of 689.1 Mb and 449,796 scaffold count was recently published. Traditionally, *L. alba* has been used for various medicinal purposes (Cardona et al. 2024), including treating digestive system ailments such as diarrhea and colic, functioning as an analgesic or antiseptic for colds, acting as an expectorant for allergies, and addressing menstrual disorders, hypertension, anemia, and skin diseases (Pedro et al. 2014; Buitrago Zapata et al. 2018; Pájaro-González et al. 2022). It is attributed with various pharmacological potentials, including antibacterial, antiviral, antifungal, antiprotozoal, neurosedative, analgesic, anti-inflammatory, antioxidant, anti-spasmodic, and cytotoxic activities depending on the method of administration, either by infusion or maceration. In Colombia, it is recognized by "Instituto Nacional de Vigilancia de Medicamentos y Alimentos" (INVIMA) in the list of medicinal plants for therapeutic purposes as an external antiseptic (Buitrago Zapata et al. 2018; INVIMA 2018; Coronel Alva and Zevallos Escobar 2019).

On the other hand, the specie *Petiveria alliacea* L. (Caryophyllales: Petiveraceae), known by common names in Spanish of "Anamú," "Chimú," "Mucura," "Mucuracá," "Palindera," "Namú," "Lancetilla," "Mapurite," or "fimercarra" (Bernal et al. 2017). It is a subshrub or shrub and grows primarily in the wet tropical biome and can be found in Colombia both in the wild and cultivated, ranging from 5 to 1780 masl. This plant maintains a "Least Concern" (LC) conservation status according to the IUCN and features on Colombia's National Red List (ColPlantA 2023). It has a maximum height of up to 1.5 m and can reproduce sexually or asexually, typically being self-pollination. Its flowers are white and arranged in terminal clusters. A total of 14 secondary metabolites have been identified, primarily encompassing alkaloids, flavonoids, triterpenes, tannins, benzoic acid, lignoceric acid, and stearic acid, among others. While the genome size remains unknown and lacks a published reference genome, its chloroplast sequence length measures 154,947 bp (Ministerio de Protección Social 2008; Alarcón Restrepo 2011; PCIDB 2023). Traditionally, it has been used to treat fever, as an antiseptic, anti-inflammatory, and analgesic for skin conditions, and anti-asthma, as an animal abortifacient. Its applications extend to treating sinusitis, arthritis, migraines, anxiety, and epilepsy, with believed effects on memory. Moreover, it is considered to inhibit the metastasis of cancer cells and is included in the list of accepted medicinal plants for therapeutic purposes as an oral mucosal anti-inflammatory (Ministerio de Protección Social 2008; INVIMA 2018; Pájaro-González et al. 2022). Regarding its pharmacological potential, *P. alliacea* is an agent that prevents tumor proliferation in prostate cancer and myeloblastic leukemia (Ruffa et al. 2002). Its extract has been studied for

its ability to induce apoptosis in 4T1 breast cancer cells and halt the growth of these cell colonies (Hernandez and Gally Jordá 1981).

Considering the scarcity of genetic information available for these two species, this study aimed to investigate the genetic diversity and population structure of the medicinal plants *L. alba* and *P. alliacea* sampled from different departments across Colombia, where the GAT program was interested in establishing crops and where the medicinal use of these plants in the wild was recorded.

Materials and Methods

Sample Collection, DNA Extraction, and RADseq Library Preparation

For the genomic analysis of these species, 20 individuals were collected for *L. alba*, five (5) were obtained from the Tolima department to 375 masl, municipality of Lérida; six (6) from the Putumayo department to 2,067 masl, municipality of Sibundoy; and nine (9) from the Valle del Cauca department to 978 masl, municipality of Palmira. These latter samples pertain to various locations in Colombia as they are part of a Germplasm Bank for medicinal plants located in the municipality of Palmira (Fig. 1). Regarding *P. alliacea*, 31 individuals were collected, 16 from the Tolima department to 230 masl, ten (10) from the municipality of Ambalema, and six (6) from the municipality of Armero Guayabal. Additionally, five (5) individuals were collected from the Cundinamarca department to 1329 masl, municipality of Pacho; five (5) from the Boyacá department to 2062 masl, municipality of Villa de Leyva; and five (5) from the Valle del Cauca department to 978 masl, municipality of Palmira (Fig. 1). Of each individual was collected between 3 and 5 leaf samples were, ensuring that these were young leaves or new plant shoots and in good condition, free from signs of predation, fungi, or pathogens. These samples were preserved in paper envelopes accompanied by silica gel to prevent decomposition or fungi growth. Additionally, of each population were deposited one sample collected in the herbarium of Pontificia Universidad Javeriana in Bogotá D.C

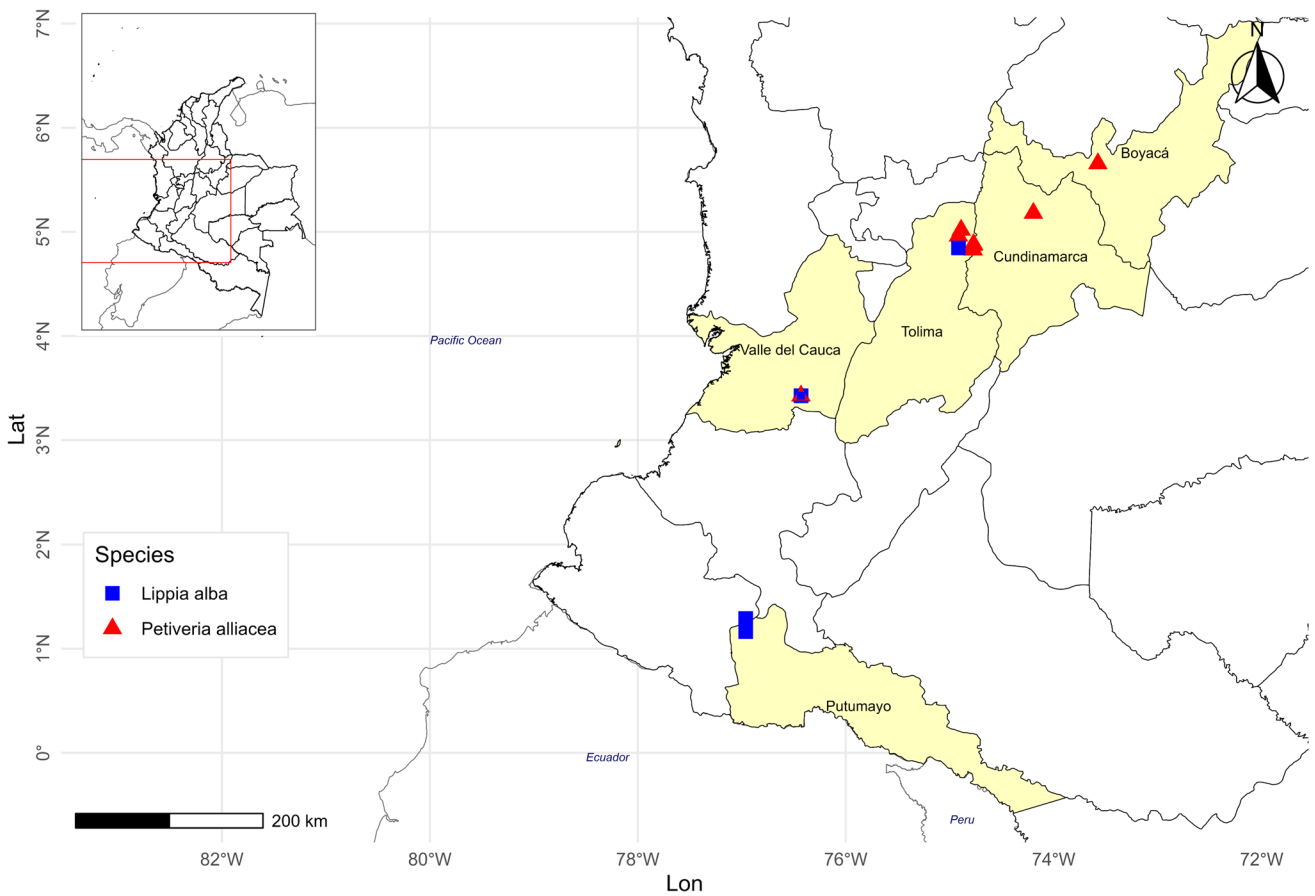


Fig. 1 Sampling points individuals *L. alba* and *P. alliacea.* Blue squares correspond to *L. alba* collected in the departments of Putumayo, Valle del Cauca y Tolima, and red triangles correspond to *P. alliacea* collected in the departments of Valle del Cauca, Tolima, Cundinamarca, and Boyacá

and José Cuatrecasas Arumí of the Universidad Nacional de Colombia in Palmira Valle del Cauca (see Supplementary Table 1).

The genomic DNA (gDNA) was isolated from 100 mg of fresh leaf tissue using Wizard® Genomic DNA Purification (PROMEGA, MD) following the manufacturer's protocol. DNA concentrations were measured by Qubit 4™ DNA Assays, Kit Qubit™ 1X dsDNA High Sensitivity (HS) Fluorometer (Thermo Fisher Scientific Inc.). The quality of gDNA samples was evaluated using Berthold Technologies Colibri microvolume spectrometer and by 2% agarose gel stained with GelRed DNA stain using the ENDURO™ Gel XL Electrophoresis System; RADseq libraries were constructed using a protocol adapted from Baird et al. (2008), through Floragenex Inc. (Beaverton, Oregon, USA) using the restriction enzyme PstI. The samples were sequenced on an Illumina® NovaSeq 6000 platform to produce 100-bp single-end reads.

Bioinformatic Analyses

Demultiplexing the Data and SNP Calling

The quality of raw reads was assessed using FastQC v 0.12.1 (Wingett and Andrews 2018). The STACKS v 2.64 software pipeline (Catchen et al. 2011) was employed for SNP calling. For the two species, denovo_map pipeline was run; additionally, for *L. alba*, ref_map was running with its reference genome. Raw reads were demultiplexed using the process_radtags script, and all samples were checked for barcode and RAD cut site integrity. Low-quality reads, defined as those with a Phred score of less than 20, were filtered out. For denovo_map, the ustacks tool uses two main parameters within the process, which affect how loci are constructed within each sample and should be tested considering different decision frameworks (see Supplementary Note 1). Then, the cstacks script generated the catalog of consensus loci by merging the alleles from each population, and comparisons were made between samples from the created catalog using the sstacks tool. Next, gstacks identified and genotyped the SNPs within the metapopulation under the variant calling model and genotype likelihoods (Maruki and Lynch 2017) with a threshold alpha ranging between 0.01 and 0.05 for each locus. The "populations" script then generated VCF (Variant Calling File) output file used for genetic and structure analysis. For pipeline running through ref_map, we aligned our samples with the reference genome of *L. alba* (ASM3783412v1) using the BWA software (Li and Durbin 2010) before running gstacks and populations tools of Stacks.

Additionally, using TASSEL v5 (Bradbury et al. 2007) filtered the VCF file of denovo_map where a locus was required to be present between at least 70 and 80% of the individuals and ref_map for *L. alba* was required that the locus to be present in 90% of the individuals.

Genetic Diversity Analyses

The genetic diversity of the natural populations was assessed by estimating observed heterozygosity (Ho), expected heterozygosity (He), transition and transversions numbers, nucleotide compositions and diversity (Pi), and number of polymorphic sites using Tajima *D* test (Tajima 1989); we evaluated the molecular indices Theta S (Θ_S) and Theta Pi ($\Theta\pi$) which represents the distribution of variation within or between populations based on polymorphic sites, values of inbreeding coefficient (F_{IS}), and HW test in Arlequin v 3.5.2.2 (Excoffier and Lischer 2010); we also analyzed the gene flow level among all populations with the Nm parameter using equation Nm = (1-Fst)/(4Fst).

Population Structure and Phylogeny

Three distinct methods were used to ascertain the genetic structure of the populations. First, ADMIXTURE v 1.3.0 (Excoffier and Lischer 2010) was employed, utilizing the maximum likelihood algorithm to explore the genetic structure of the data. Ten independent runs were performed for each K value, from 1 to 5 for *L. alba* and from 1 to 6 for *P. alliacea*. To visualize the optimal number of delta *K* values and the model of population structure, the .Q and .P files were used as input in the software STRUCTURE SELECTOR (Li and Liu 2018), along with a population map. Second, we performed a principal component analysis (PCA) using the SNPrelate v 1.34.1. (Zheng et al. 2012) package and ggplot2 v 3.4.2 (Wickham 2009) in R 4.3.2 (R Core Team, 2021) to extract the synthetic variables to represent major genetic variations. Thirdly, a maximum likelihood (ML) phylogenetic inference was performed using IQ-TREE v 2.1.4 (Minh et al. 2020) with 150,000 bootstrap replicates. The tree was visualized using iTOL v 5 (Letunic and Bork 2021) and the substitution model was determined using the software jModelTest v 2.1.10 (Darriba et al. 2012) employing the Bayesian model selection criterion with support from 24 models provided by default. To evaluate the correlation between the geographical distance and the genetic distance of the populations, a Mantel test was performed using the software IBD (Isolation by Distance) (Bohonak 2002) and an AMOVA test considering Wright's population structure statistics in Arlequin (Excoffier and Lischer 2010).

Finally, an FST outlier approach was used to determine a set of candidate SNPs that had significantly higher FST values than expected under a neutral model of selection using the FDIST (Beaumont and Nichols 1996) implemented in Arlequin. This method detects genes under selection based on the distributions of heterozygosity and FST and can

be used to determine SNPs that are likely under divergent selection, whereby knowing their location the association of these with a functional gene could be known (Excoffier and Lischer 2010). We selected SNPs that had significantly higher FST values than expected; we ran 10,000 simulations with 100 demes per species and minimum and maximum expected heterozygosities of 0 and 0.8, respectively. We considered markers with FST values higher than 95% (> 0.05) of the neutral distribution were considered to be under divergent selection, and markers with FST lower than 95% of neutral distribution were inferred to be subject to balancing selection (Beaumont and Nichols 1996). Once the SNPs identified under divergent selection were obtained, a BLAST (Altschul et al. 1990) was performed in NCBI and UniProt (The UniProt Consortium 2021) to obtain relevant information about possible genes or proteins that were related to them.

Results

RADseq Libraries and SNP Calling

After sequencing the RAD libraries of the 20 samples from *L. alba*, we obtained 3.8 GB of raw data consisting of more than 25 million reads measuring 123 bp. The raw data showed high Phred quality Q30 for more than 90% of the samples with an average GC content of 46% (sd. 4.6), and for *P. alliacea*, we obtained 16 GB of raw data from 31 samples, consisting more of 163 million reads with high Phred quality (Q30 > 90%), 123 bp in length, and an average GC content of 42% (sd. 4.0).

For denovo_map assembly, once we established the parameters for ustacks, ran each component individually. For the *L. alba*, we obtained 78,753,775 RADtags (Supplementary Table 2) that consisted of 2,704,413 RAD loci with an effective per sample mean coverage of 19.3x (min = 14.7x, max = 30.8x), composed of 404,517 variant sites, 196,585 SNPs, and an average of 41,175 private alleles per population (Table 1). After removing SNPs with missing data in TASEEL, 17,036 SNPs were retained in 16 samples. For *P. alliacea* after assembly, 325,195,434 RADtags were obtained (Supplementary Table 2), forming 1,667,988 RAD loci with an effective per-sample mean coverage of 96.2x (min = 10.5x, max = 445.6x), composed of 172,199 variant sites, 19,491 SNPs, and an average of 17,950 private alleles per population (Table 1). After removing SNPs with missing data in TASSEL, 6395 SNPs were retained in 25 samples. In the results of the type of mutation that the populations show, it was evident that more than 50% of the SNPs at the population level were transitioned for the species *L. alba*, while for *P. alliacea*, the Tolima population had 55% transversions, whereas for the other populations, the SNPs were transitions in 60% (Fig. 2; Table 2). On the other hand, in ref_map assembly for *L. alba*, we obtained forming 478,304 RAD loci with an effective per-sample mean coverage of 22.4x (min = 10.0x, max = 49.2.6x), composed of 309,642 variant sites, 49,874 SNPs, that after filtering with TASSEL, where required that the variations were present in 90% of samples of populations obtained 14,562 SNPs were retained in 17 samples; most of these (10,241) are located in the first 11 scaffolds of the genome annotation (Supplementary Fig. 1). Despite having retained more SNPs for the *L. alba* populations, the behavior in the results of genetic diversity and population structure for this analysis had the same trend as that obtained in denovo_map (see Supplementary Note 2).

Genetic Diversity

Using the data obtained after filtering from the polymorphic sites, we calculated genetic variation for the three evaluated populations of *L. alba* and four populations of *P. alliacea*. For the first species, 17,036 of the total SNPs filtered were useful for the diversity analyses. Similar values of genomic diversity were found among populations, with Ho mean of 0.225 ± 0.171 (Valle del Cauca), 0.371 ± 0.225 (Tolima), and

Table 1 Population summary for *L. alba* and *P. alliacea* after assembly in Stacks v 2.64

	Population	No. of individuals	Total No. of sites	Variants	Total No. of SNPs
L. alba	Tolima	5	37,828,498	246,037	137,055
	Putumayo	6	121,191,251	278,400	182,804
	Valle del Cauca	9	214,574,808	360,511	269,896
P. alliacea	Cundinamarca	5	62,479,731	119,079	57,091
	Tolima	16	46,103,330	124,114	21,916
			19,438,200	108,079	16,983
			41,219,323	120,788	11,874
			43,424,457	120,188	12,625
	Boyacá	5	12,338,289	93,021	7763
	Valle del Cauca	5	33,352,257	83,272	8183

Table 2 Summarized genetic diversity statistics for populations of *L. alba* and *P. alliacea*

Species	*L. alba*			*P. alliacea*			
Population	Tolima	Putumayo	Valle del Cauca	Cundinamarca	Tolima	Boyacá	Valle del Cauca
No. of usable loci	8563	8695	5347	4037	3729	3119	3984
No. of ts. sites	2733	2730	2459	1270	318	137	105
No. of tv. sites	1839	1892	1586	882	389	95	95
SNPs	4572	4622	4045	2152	707	232	200
Ho	0.371	0.416	0.16	0.553	0.091	0.301	0.555
He	0.396	0.397	0.24	0.499	0.159	0.379	0.620
Pi	0.201 ± 0.106	0.200 ± 0.106	0.228 ± 0.118	0.347 ± 0.189	0.029 ± 0.014	0.026 ± 0.014	0.027 ± 0.018
GC%	53.1	53	52.7	49.9	50.5	51.1	50.7
Theta S (Θ_S)	1616 ± 649	1633 ± 656	1339 ± 512	829 ± 355	178 ± 54	89 ± 38	109 ± 58
Theha Pi (Θ_π)	1725 ± 911	1745 ± 922	1219 ± 631	1402 ± 765	111 ± 54	84 ± 46	110 ± 72
Fis	0.07	−0.06	0.35	−0.12	0.43	0.24	0.20

No. of ts. sites transitions sites, *No. of tv. sites* transversions sites, *SNPs* number of polymorphic sites, *Na* number of alleles, *Ho* observed heterozygosity, *He* expected heterozygosity, *Pi* nucleotidic diversity, *GC%* percent Guanine Cytosine, *ΘS* Theta S, *Θπ* Theta Pi, *Fis* inbreeding coefficient

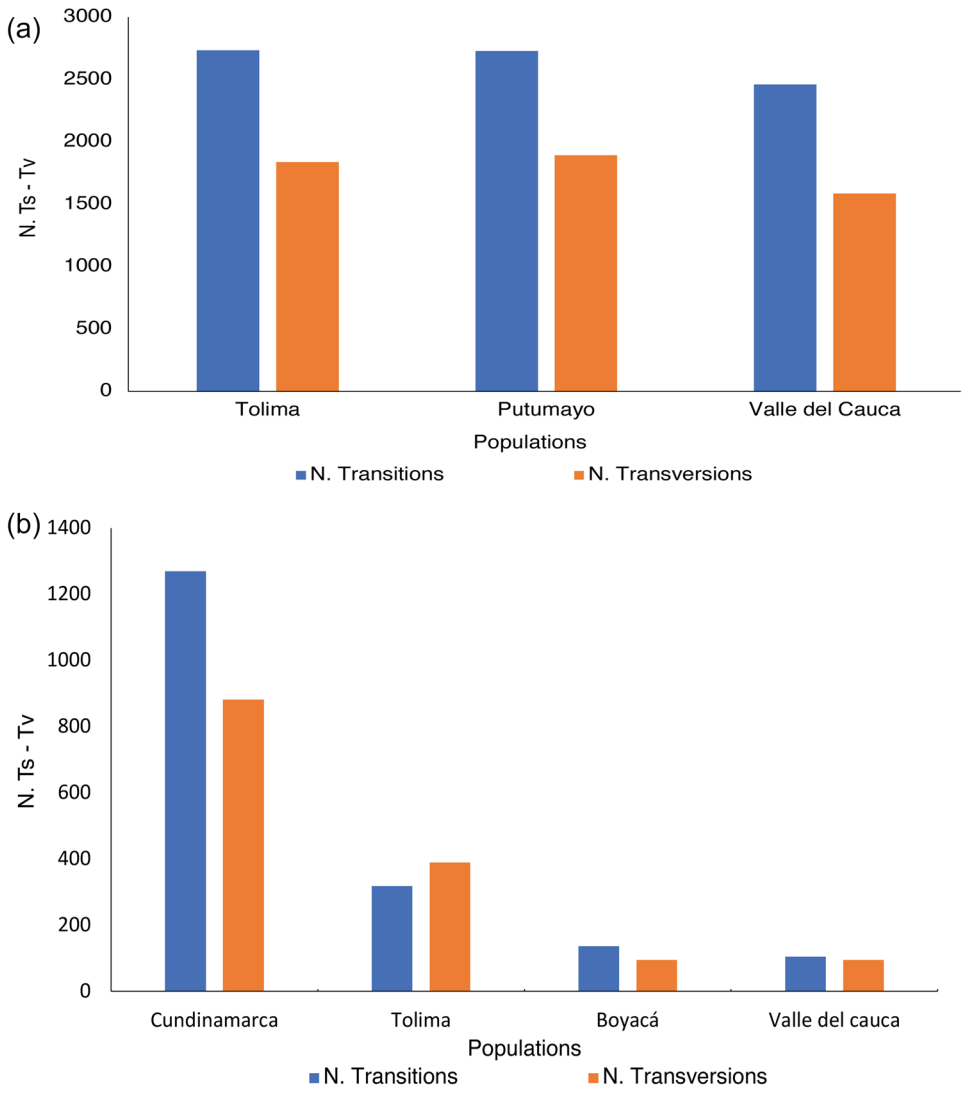


Fig. 2 *Number of transitions vs transversions*. Mutation type of the SNP for the seven populations. **a** Number of transitions and transversions for the three populations of *L. alba*. **b** Number of transitions and transversions for the four populations of *P. alliacea*

Table 3 Analyses of molecular variance (AMOVA) for *L. alba* and *P. alliacea*

Species	Source of variation	Sum of squares	Variance components	Percentage of variation
L. alba	Among populations	5771.424	69.301	3.40
	Among individuals within populations	24,277.517	290.535	14.3
	Within individuals	23,143.500	1678.180	82.3
P. alliacea	Among populations	14,254.584	537.839	75
	Among individuals within populations	3660.728	10.369	1.4
	Within individuals	3953.500	170.954	23.6

0.416 ± 0.238 (Putumayo) and He of 0.339 ± 0.135 (Valle del Cauca), 0.396 ± 0.132 (Tolima), and 0.397 ± 0.131 (Putumayo). The nucleotide diversity average over loci was 0.228 ± 0.117 (Valle del Cauca), 0.201 ± 0.105 (Tolima), and 0.200 ± 0.105 (Putumayo); molecular diversity Θπ was higher than ΘS; Fis value was negative for the Putumayo, which is congruent with the Ho value; and the *p*-value = 0 of the HW test, meaning that there is an excess of heterozygotes in this population (Table 2). For *P. alliacea*, 6395 SNPs filtered were useful for the diversity analyses. These populations had a Ho mean of 0.553 ± 0.241 (Cundinamarca), 0.091 ± 0.163 (Tolima), 0.301 ± 0.225 (Boyacá), and 0.555 ± 0.298 (Valle del Cauca) and He of 0.499 ± 0.105 (Cundinamarca), 0.159 ± 0.154 (Tolima), 0.379 ± 0.126 (Boyacá), and 0.620 ± 0.187 (Valle del Cauca). The nucleotide diversity average over loci was 0.261 ± 0.141 (Cundinamarca), 0.029 ± 0.014 (Tolima), 0.026 ± 0.014 (Boyacá), and 0.027 ± 0.017 (Valle del Cauca); the molecular diversity ΘS was higher than Θπ; and Fis value has the same behavior as in *L. alba*, being negative for Cundinamarca, where the Ho value is greater than He and the *p*-value = 0 of the HW test (Table 2).

Genetic Differentiation and Analysis of Molecular Variance

The population structure was analyzed by an AMOVA. For *L. alba*, we found that 3.4% of the molecular variation was distributed among populations, while 82.3% from within populations; the fixation indices were FST 0.034 ($p < 0.001$), FIS 0.147 ($p < 0.001$), and FIT 0.176 ($p < 0.001$) (Table 3), with a marked differentiation between populations from Tolima and Putumayo of 0.062 $p < 0.001$ (Table 4); the Nm parameter was 5 and the Mantel correlation test was $r = 0.964$ ($p = 0.159$) indicating a possible correlation between genetic (FST) and geographical distances between populations. On the other hand, the hierarchical AMOVA for *P. alliacea* showed that 75% of the variance was attributed among populations and 23.6% within individuals; the fixation indices were FST 0.747 ($p < 0.001$), FIS 0.057 ($p < 0.001$), and FIT 0.762 ($p < 0.001$) (Table 3). Genetic differentiation values between population pairs were significant in all comparisons except for those between Cundinamarca and Valle del Cauca (Table 5); the Nm parameter was 0 and the Mantel correlation test was $r = 0.439$ ($p = 0.238$) indicating a lack of correlation between genetic and geographical distances between populations.

Population Structure

The ADMIXTURE population structure analysis of the evaluated species revealed patterns similar to the genetic diversity analyses. For the species *L. alba*, the three evaluated populations were divided into two optimal genetic groups with a CV of K2: 0.845 (Fig. 3; Supplementary Fig. 2), with some individuals showing a mixture of clusters such as Tol-2, Tol-4, VII_1, VII_4, VII_5, and VII_7. However, all individuals from Putumayo, Tol-3, Tol-5, and VII-3 recovered within the same genetic group (Supplementary Table 3). The

Table 4 Distance method: pairwise differences for populations of *L. alba* FST are below the diagonal, while significance values ($p < 0.001$) are shown on the diagonal

Population	Tolima	Putumayo	Valle del Cauca
Tolima		*	-
Putumayo	0.062		-
Valle del Cauca	0.011	0.029	

Asterisk (*) means significantly different; hyphen minus (-) means not significant

Table 5 Distance method: pairwise differences for populations of *P. alliacea* FST are below the diagonal, while significance values ($p < 0.001$) are below the diagonal

Population	Cundinamarca	Tolima	Boyacá	Valle del Cauca
Cundinamarca		*	*	*
Tolima	0.685		*	*
Boyacá	0.575	0.830		*
Valle del Cauca	0.510	0.847	0.877	

Asterisk (*) means significantly different

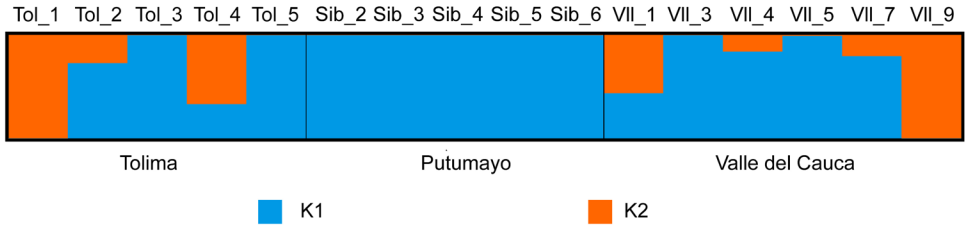


Fig. 3 Distribution of genetic groups in *L. alba* populations given by ADMIXTURE. Clustering of 16 individuals of *L. alba* based on Admixture estimation for $K=2$. Each bar indicates an individual while the colors correspond to membership to a particular cluster; the number of clusters is determined by the K value, the color blue corresponds to K1, and orange is K2

PCA also showed the same trend (Fig. 4; Supplementary Fig. 3), where the first component explained 12.1% of the variation, separating all individuals from Tolima and three individuals from Valle del Cauca (Vll_1, VII_3, and VII_9) vs all individuals from Putumayo and three individuals from Valle del Cauca (VII_4, VII_5, and VII_7). In contrast, the second component explained 9.9% of the variation separating individual VII_1 from Valle del Cauca from the rest samples. The maximum likelihood phylogeny showed the same trend as the other analyses; the optimal model given by jModelTest (Darriba et al. 2012) was GTR + G with a gamma value of 0.903, revealing two clades each with a bootstrap robustness value of 100 (Fig. 5). Within these clades, there appears to be a possible relationship between individuals from the Tolima and Valle departments (VII_1, VII_3, and VII_9), and it is also possible that individuals VII_4, VII_5, and VII_7 are related to those from Putumayo.

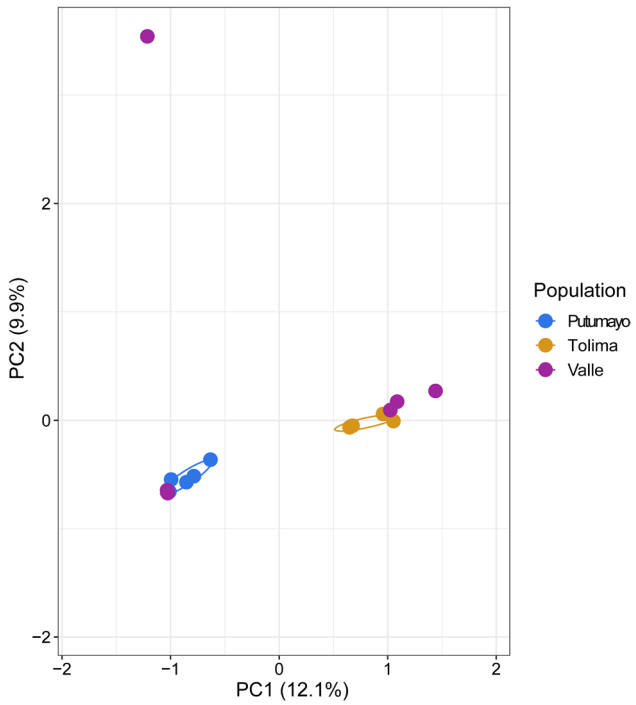


Fig. 4 PCA for *L. alba* populations. PCA plot showing the genetic differentiation among the three natural populations of *L. alba* analyzed. Purple dots represent the Valle del Cauca individuals, blue dots represent Putumayo individuals, and orange dots represent Tolima individuals

On the other hand, the ADMIXTURE analysis conducted for *P. alliacea* established $K=4$ as the optimal grouping with a CV = 0.313 (Fig. 6; Supplementary Fig. 4). In this grouping, all sampling sites from the Tolima department cluster together, while all from Boyacá cluster in another. Samples from the Cundinamarca department showed an average of 29% genomic similarity with those from Valle del Cauca (Supplementary Table 4) despite each corresponding to a defined genetic group. The PCA also exhibited the same trend, with samples from each department grouped together; the first component represented 29.6% of the variation, separating all samples from Cundinamarca vs all others samples, while the second component explained 18.3% (Fig. 7; Supplementary Fig. 5), separating samples from Boyacá and Valle del Cauca from those in Cundinamarca and Tolima. The maximum likelihood phylogeny showed a similar pattern to the other analyses; the optimal model given by jModelTest (Darriba et al. 2012) was GTR + G with a gamma value of 0.899, revealing a division into three clades, each supported with a bootstrap value of 100 (Fig. 8). The first clade contained all the individuals from Tolima, the second clade included all samples from Cundinamarca, and a third clade consisted of samples from Boyacá and Valle del Cauca. Cundinamarca and a third clade consisted of samples from Boyacá and Valle del Cauca. Within this third clade, substructuring was apparent, with individuals from each department separated into subclades, each supported by a bootstrap value of 100.

Lastly, in the island model (FDIST) obtained in Arlequin conducted to detect loci under selection taking into account the FST value and the observed heterozygosity, we observed that for *P. alliacea*, 2276 SNPs were classified under alternative selection pressure conditions. Of these, 796 SNPs showed high values ($FST > 0.74$, $p < 0.001$), suggesting they could be under divergent selection (Fig. 9); we attempted to map them by gene orthology to conduct association analyses with some functional genes related to metabolite production. However, we found that this mapping is not possible without prior annotation.

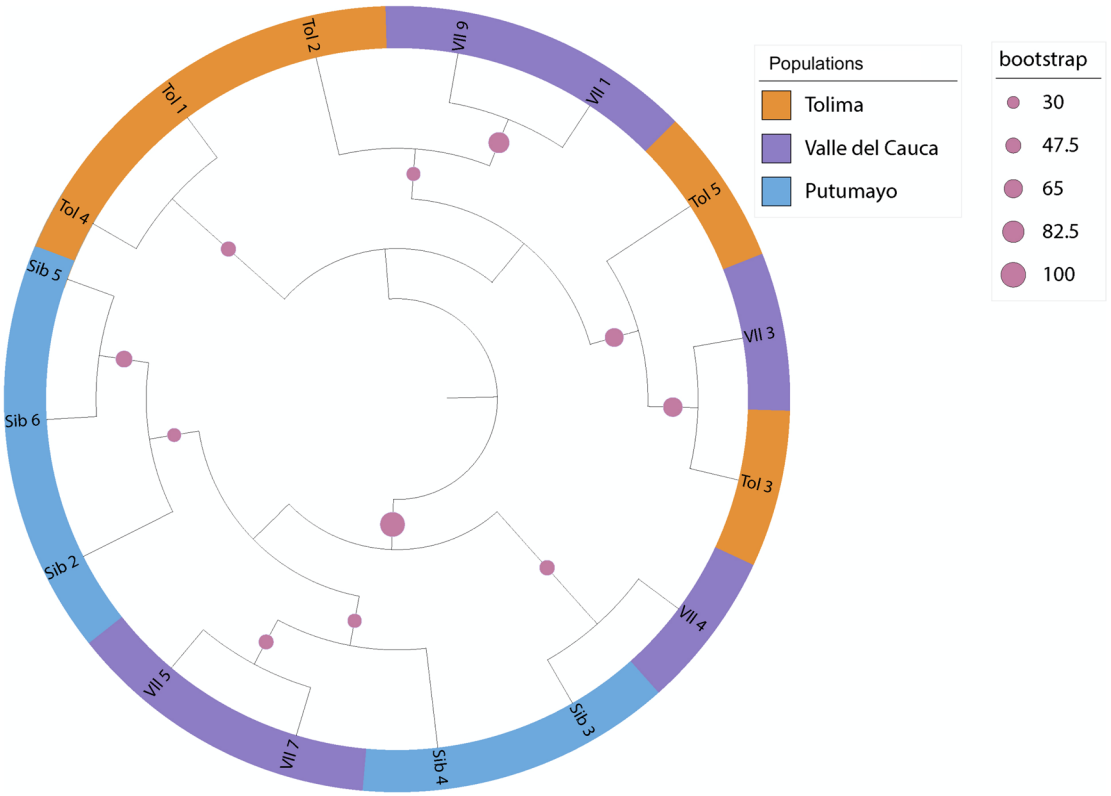


Fig. 5 Phylogenetic tree based of natural population of *L. alba.* Phylogenetic tree inferred from maximum likelihood based on the analysis of 16 individuals of *L. alba*. Purple branches represent the Valle del Cauca individuals, blue branches represent Putumayo individuals, and orange branches represent Tolima individuals

Therefore, for *L. alba*, we use the FDIST distribution obtained in the ref_map pipeline, where of 14,563 SNPs, 4081 SNPs were classified under conditions of alternative selection pressure (Fig. 10). Among them, 705 SNPs showed high values (FST > 0.04, $p < 0.001$) (Supplementary Note 2), suggesting they might be under divergent selection, while the remaining 3184 SNPs showed low FST values (FST < 0.04, $p < 0.001$), which might indicate balancing selection. From these 705 SNPs, we extracted the sequences of each one from the population.loci.fa file generated in stacks, to perform a BLASTn (Supplementary Table 5), where we limited the search in the nt database to organisms belonging to the order Lamiales. Once this BLAST was executed, we obtained 200 sequences, of which 37 belonged to proteins that could encode a gene of interest; the rest matched with a chromosome in the annotation of the genome of different plants. These 37 sequences were BLASTed using the UniProt database to obtain possible associations with genes of

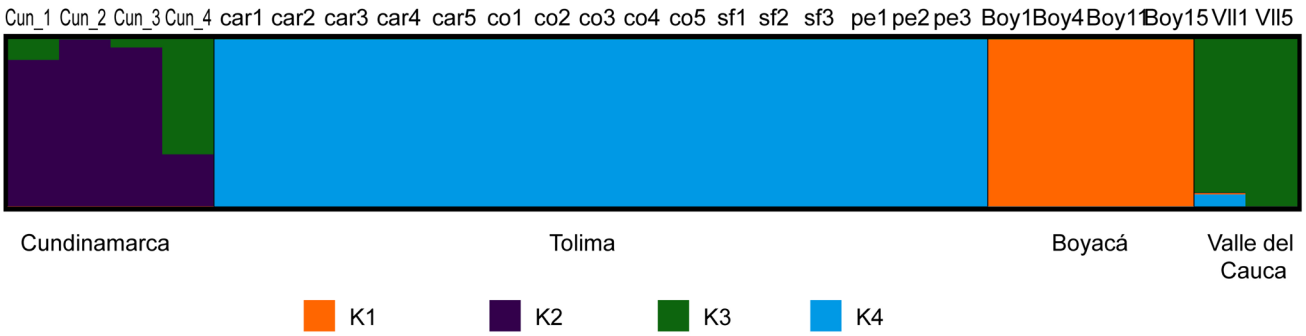


Fig. 6 Distribution of genetic groups in *P. alliacea* populations given by ADMIXTURE. Clustering of 25 individuals of *P. alliacea* based on the Admixture estimation for $K = 4$. Each bar indicates an individual and each color a membership to a particular cluster; the number of clusters is determined by the *K* value, orange color corresponding to K1, purple K2, green K3, and blue K4

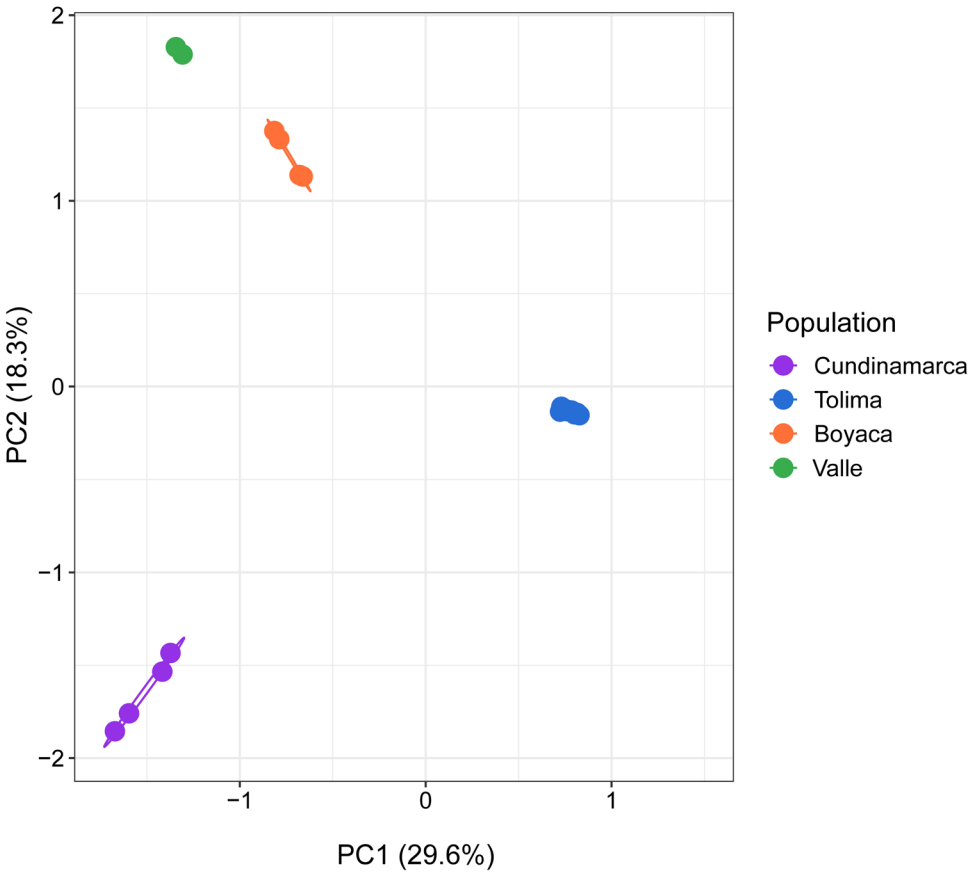


Fig. 7 PCA for *P. alliacea* populations. PCA plot showing the genetic relationships among the four natural populations of *P. alliacea* analyzed. The purple dots represent the Cundinamarca individuals, the blue dots represent Tolima individuals, the green dots represent Valle del Cauca individuals, and the yellow dots represent Boyacá individuals

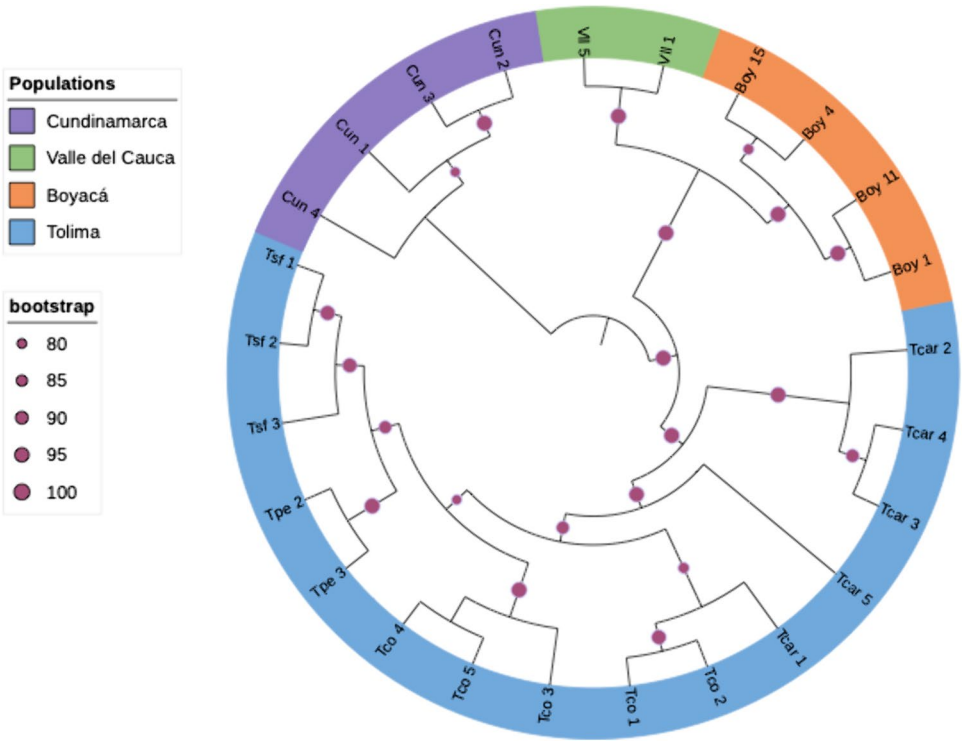


Fig. 8 Phylogenetic tree of natural population of *P. alliacea*. Phylogenetic tree inferred from maximum likelihood based on the analysis of 25 individuals of *P. alliacea*.

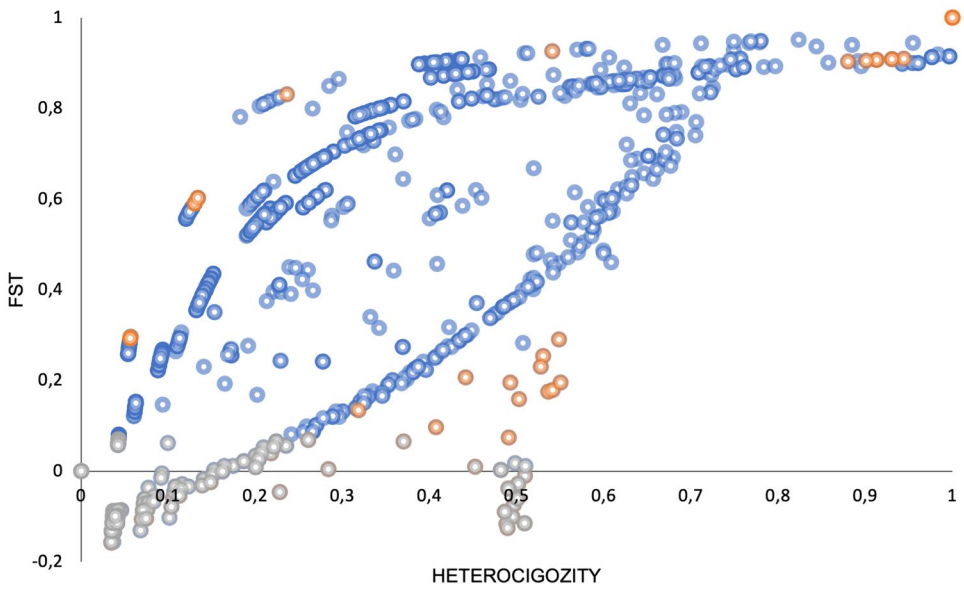


Fig. 9 Island model for *P. alliacea*. FDIST for 6895 SNPs of *P. alliacea*. SNPs under divergent selection (796, orange points), balancing selection (1480, grey points), and 619 under neutral selection (blue points)

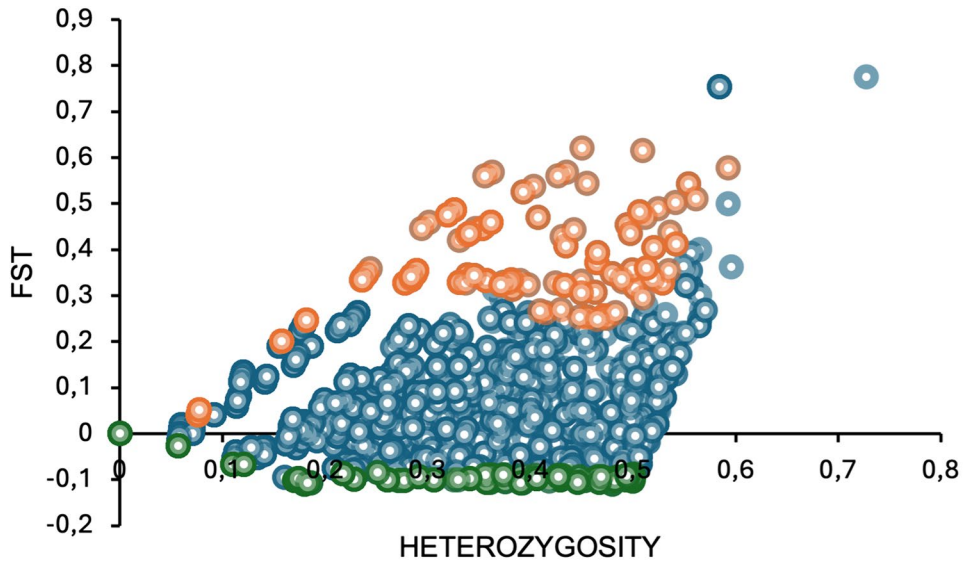


Fig. 10 Island model for *L. alba*. FDIST for 14,562 SNPs of *L. alba*. SNPs under divergent selection (705, orange points), balancing selection (3184, green points), and 10,674 under neutral selection (blue points)

interest (Supplementary Table 6), where we identified the biological and molecular functions and cellular components of each one of them. In general, the SNPs identified belong to chromosomes and mRNA processes. Among the genes found, we highlight *SASPL_104284*, which is part of the metabolic process in the production of terpene in the specie *Salvia splendens*.

Discussion

We consider the results here to be the first record of the genetic diversity and population structure of these two species in Colombia using RADseq methodology. The number of retained reads, loci, and SNPs found in this study are consistent with results reported in other population studies of non-model plants utilizing similar methodologies (Clugston et al. 2019; Palombo and Carrizo García 2022). Furthermore, this study demonstrated that the use of next-generation sequencing methodologies such as RADseq allows for the generation of a high number of single nucleotide polymorphisms, thereby improving genetic analyses. Our study provides evidence supporting the efficiency of Restriction-site Associated DNA sequencing (RADseq) in reliably identifying SNPs. The sets of SNPs that we identified proved useful in analyzing the genetic diversity and population structure of the two evaluated medicinal plant species. For the species *L. alba*, a division into two genetic groups was observed

across the three evaluated populations, displaying notably higher genetic diversity within these individual populations, for denovo_map and ref_map analysis accounting for approximately 94.8% of the overall variation. Samples from the Tolima and Valle del Cauca populations exhibited potential genetic relatedness, possibly related to the plant's reproductive modes involving both seeds and cuttings. In contrast, the samples sourced from Putumayo have been propagated via cuttings, potentially impacting the observed population structure evident in our findings. This behavior in the population structure aligns with the molecular indices, notably akin to the populations. Theta Pi values, representing the average number of nucleotide differences per pair, surpassed Theta *S* values, based on polymorphic sites, for three populations, which corroborates our results by showing greater genetic variation within the population. Additionally, the He values slightly surpassed Ho values, indicating no significant genetic distinctions among these populations. However, a noteworthy and significant genetic distance was discerned between the Tolima and Putumayo populations, accompanied by a higher Ho. Despite this, the nucleotide diversity patterns remained consistent across all populations, with relatively elevated nucleotide transition values. According to Gao et al. (2017), these patterns might be linked to biased mutational processes inherent within plant genomes, for instance, postmeiotic abnormalities affect the formation of viable pollen grains, leading to a significant decrease in *L. alba* seed germination, thereby posing challenges to its sexual reproduction (Kaur and Singhal 2019). Additionally, there is a unique mode of dissemination where adults from cultural and ethnic groups contribute to dispersal, given its widespread medicinal use and ease of propagation through cuttings or stakes (Herrera-Moreno et al. 2013).On the other hand, for the values Ho higher than He, we can relate them to the opposite phenomenon of the Wahlund effect, which consists of the so-called isolation breakage, where a temporary excess of Ho can occur when geographically isolated populations reproduce (Smith et al. 2020), which can be explained for the *L. alba* samples. Additionally, the *p*-value of the Hardy–Weinberg equilibrium analysis showed that neither population of the two species is in equilibrium, and the Fis value corroborates the excess of heterozygotes for the Putumayo population for *L. alba* and the Cundinamarca population for *P. alliacea.*

Previous studies have investigated the genetic diversity of *L. alba*, utilizing diverse molecular markers. In line with our findings, these studies have reported substantial genetic variation among individuals. Paz Perafán et al. (2019) employed genotyping by GBS and identified 37,917 SNPs, attributing 93.35% of the variation to differences among individuals. Cardona et al. (2019) investigated two ecologically distinct populations (Chicamocha Canyon and the Sumapaz Paramo) using randomly amplified microsatellites (RAMS), observing the highest genetic variation within the assessed populations. Similarly, Julião et al. (2020) analyzed species of *Lippia* genus using AFLP markers and detected 273 polymorphic loci, delineating three genetic groups among seven studied populations.

In this sense, consideration of the high genetic and chemical variability observed in *L. alba* is essential for establishing conservation plans or germplasm collections that can preserve and represent the variability within this species and be used as a valuable tool for selecting genotypes of interest and reproducing new populations (Manica-Cattani et al. 2009). Furthermore, additional studies focusing on the genetic diversity of *L. alba* under diverse conditions and within various ecosystems are imperative. Similar congeneric species like *L. origanoides* have exhibited genetic variation influenced by factors such as altitude, sampling locations, and varying degrees of human impact on their reproductive processes (Vega-Vela and Sánchez 2012). Consequently, exploring these aspects in future studies for *L. alba* could yield essential insights into its adaptive mechanisms and aid in designing effective conservation and management strategies for this valuable species.

The species *P. alliacea* displayed notable genetic variation, amounting to 75%, observed primarily between populations. Our analysis revealed a robust population structure comprising four distinct genetic groups, indicating limited gene flow between these populations. The Cundinamarca population exhibited a higher count of SNPs and a greater Theta Pi value compared to the other three evaluated populations. However, despite these differences, nucleotide diversity and composition remained consistent across all four populations. In instances where Ho (observed heterozygosity) exceeded He (expected heterozygosity), this discrepancy was observed in the samples from the Cundinamarca and Valle del Cauca populations. Significant genetic distance among the four populations reaffirmed the previously identified population structure. Although studies on the genetics of this species remain limited, other species within the closely related family Phytolaccaceae have recorded polymorphic loci. For instance, using molecular markers such as RAPD, Semagn et al. (2001) estimated a total variance of 17.2% between populations, accompanied by FST values ranging from 0.350 to 0.384. These values signify substantial genetic differentiation between populations, echoing the findings observed in our study among the four evaluated populations.

Finally, we can observe that the natural variation, including the genetic variation found in these populations, may lead to various characteristics and genotypic expressions. The SNPs under divergent selection, as classified in the island analysis (FDIST) performed for both species, could be associated with different functional genes involved in the production of secondary metabolites, which confer the pharmacological potential attributed to each of these species.

We also evidenced that it is necessary to have a reference genome to be able to make an association analysis with the SNPs selected under divergent selection; in the analysis performed for the species *L. alba*, we also showed that in addition to having an annotation, this should be at the chromosomal level to make a more robust association; however, at a general level, the blast allowed us to identify genes that could be related to biological, molecular, and cellular processes in the species.

Conclusions

This study revealed genetic differentiation within the studied populations of *L. alba*, particularly within the populations from Tolima and Valle del Cauca, when compared to the Putumayo population. The reduced diversity and distinct population structure observed in Putumayo might be attributed to this species' propagation via cuttings in that region. Conversely, *P. alliacea* displayed a contrasting pattern, showcasing greater genetic variation among populations. This difference in behavior could potentially be linked to the distinct reproductive modes of these species: *L. alba* predominantly through cuttings in a single population versus *P. alliacea* potentially through seeds within populations.

Moreover, the employed methodology for de novo assembly successfully facilitated the acquisition of genetic diversity and population structure values for the studied populations. These values not only corroborate existing limited information about these species but also introduce a significant number of additional SNPs, substantially expanding the genomic resources available for these species. These expanded resources will enable more comprehensive analyses of genetic diversity and its associations with the production of secondary metabolites responsible for their recognized pharmacological potential. Lastly, this study could serve as a pioneering effort in SNP identification for these two species through the RADseq sequencing methodology.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s11105-024-01484-z.

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Availability of Data and Materials The raw RAD sequence data for the 17 samples of *L. alba* and 31 samples of *P. alliacea* have been deposited in the National Center for Biotechnology Information (NCBI) Bioproject under the accession number PRJNA1025911 and the samples among SAMN37724680 to SAMN37724727 number.

Declarations

Ethical Approval Not applicable.

Conflict of Interest The authors declare no competing interests.

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